香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

系统枢纽
SYSTEMS HUB

智能交通学域
INTELLIGENT TRANSPORTATION THRUST

LiDAR-Visual Navigation

Graduate Course INTR-6000P

Week 9 - Lecture 18

Changhao Chen

Assistant Professor

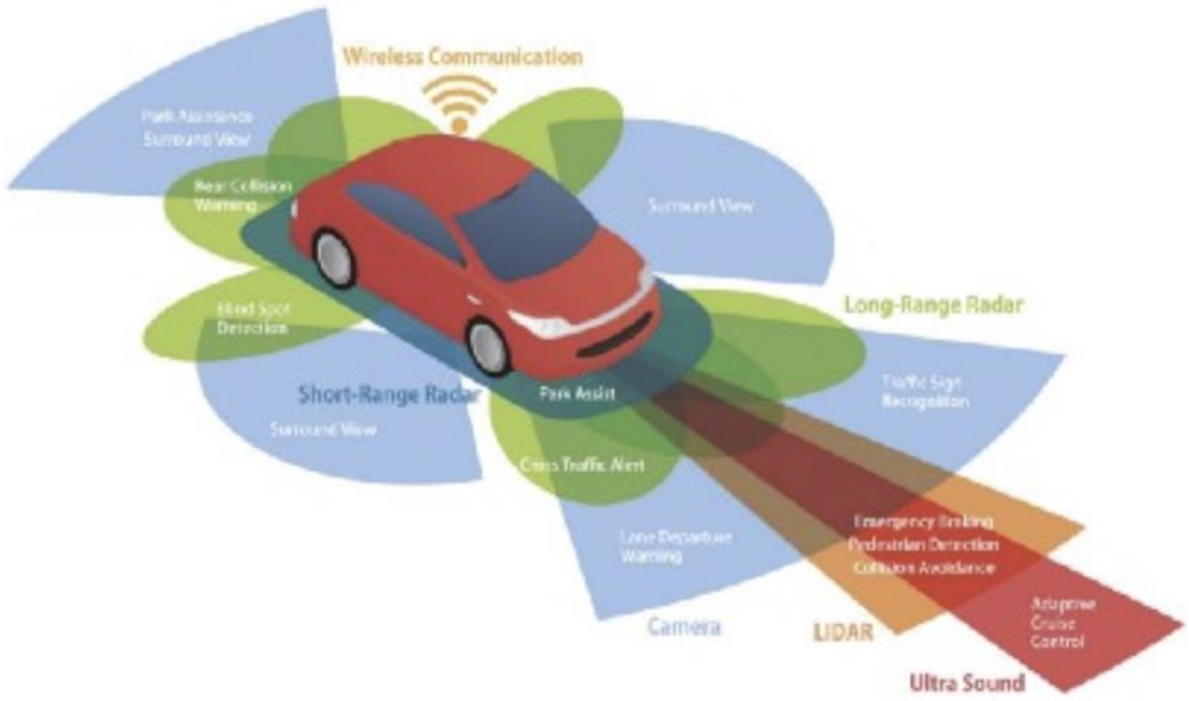
HKUST (GZ)

Motivation

LiDAR → accurate geometry & range
Camera → semantics, texture, robust loop closure
IMU → high-frequency motion constraint

Why fusion?
•Illumination + texture changes
•Range limits / degeneracy
•Cross-modal redundancy → robustness

Challenges
•Sensor synchronization
•Extrinsic calibration (LiDAR-Cam-IMU)
•Motion distortion (rotating LiDAR)
•Dynamic objects
•Real-time global consistency

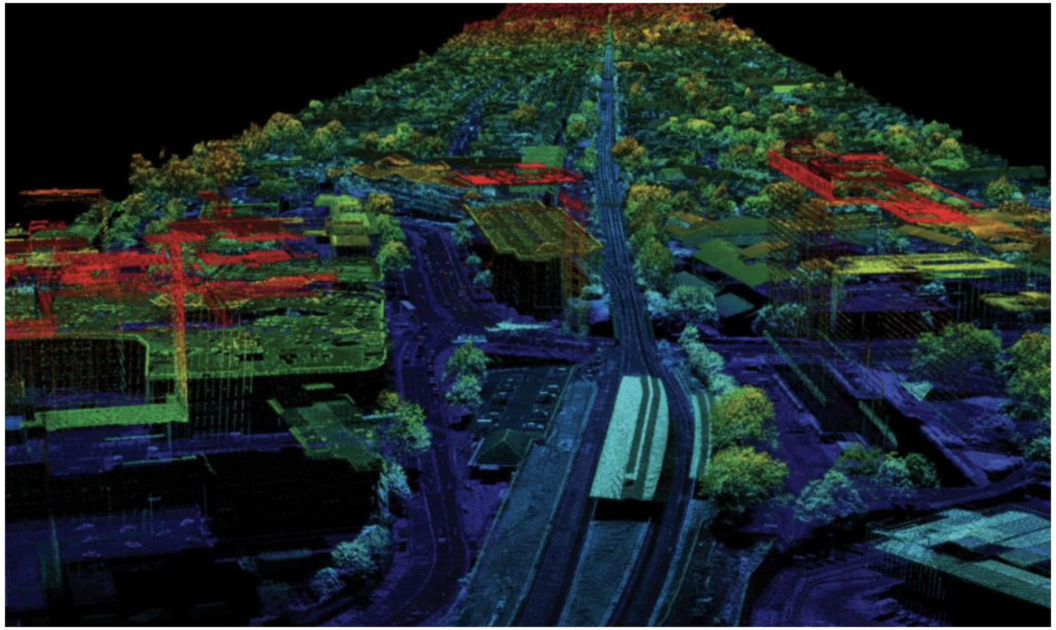


LiDAR-Visual Navigation

LiDAR-Visual Fusion Architecture

Block diagram:

•Sensors → Feature Extraction → Multi-Modal Fusion → Odometry → Mapping → Loop Closure → Global Optimization



Sensor Models

•Camera: projection, epipolar geometry

•LiDAR: SE(3) scan motion compensation

•IMU: Pre-integration

$$\mathbf{T}_{k+1} = \exp(\Delta\xi) \cdot \mathbf{T}_k$$

LiDAR Odometry

Goal: Estimate ego-motion by aligning LiDAR scans → build accurate trajectory & local map.

•Scan-to-scan / scan-to-map

Strategy	Description	Use Case
Scan-to-Scan	Align consecutive scans	Fast motion update, low latency
Scan-to-Map	Align current scan to local map	Higher accuracy, drift reduction

ICP Formulations

•**Point-to-Point ICP:** minimize $\|Rp + t - q\|^2$

•**Point-to-Plane ICP (preferred):** minimize $(n^T(Rp + t - q))^2$

→ Faster convergence using **surface normals**

Normal Constraints

•Normals estimated via local neighborhood

•Normal-consistency improves correspondence reliability

•Reject mismatches & dynamic points

Typical System 1 — LOAM

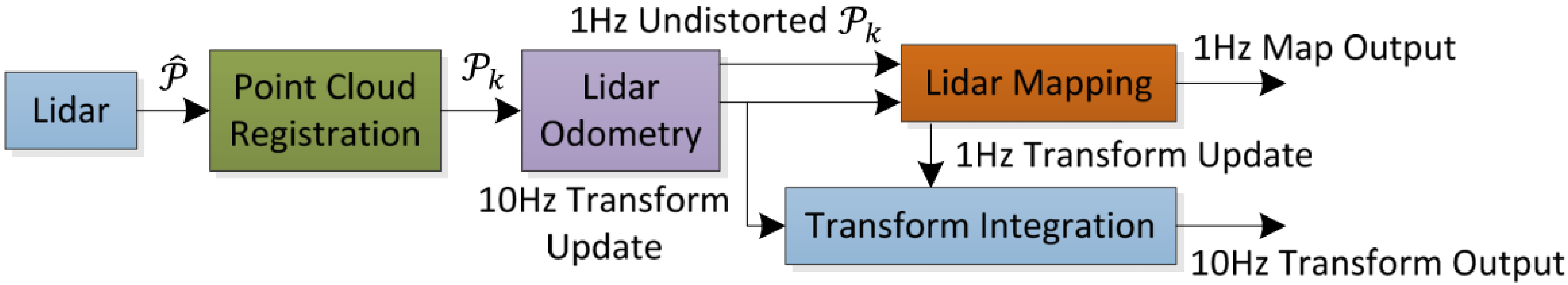
LOAM: Lidar Odometry and Mapping (2014)

Two-thread pipeline

•High-freq LiDAR odometry

•Low-freq mapping optimization

Key innovation: feature-based LiDAR



	Method	Setting	Code	Translation	Rotation	Runtime	Environment
1	SOFT2			0.53 %	0.0009 [deg/m]	0.1 s	4 cores @ 2.5 Ghz (C/C++)
I. Cvišić, I. Marković and I. Petrović: SOFT2: Stereo Visual Odometry for Road Vehicles Based on a Point-to-Epipolar-Line Metric. IEEE Transactions on Robotics 2022. I. Cvišić, I. Marković and I. Petrović: Enhanced calibration of camera setups for high-performance visual odometry. Robotics and Autonomous Systems 2022. I. Cvišić, I. Marković and I. Petrović: Recalibrating the KITTI Dataset Camera Setup for Improved Odometry Accuracy. European Conference on Mobile Robots (ECMR) 2021.							
2	V-LOAM			0.54 %	0.0013 [deg/m]	0.1 s	2 cores @ 2.5 Ghz (C/C++)
J. Zhang and S. Singh: Visual-lidar Odometry and Mapping: Low drift, Robust, and Fast. IEEE International Conference on Robotics and Automation(ICRA) 2015.							
3	LOAM			0.55 %	0.0013 [deg/m]	0.1 s	2 cores @ 2.5 Ghz (C/C++)
J. Zhang and S. Singh: LOAM: Lidar Odometry and Mapping in Real- time. Robotics: Science and Systems Conference (RSS) 2014.							

Typical System 1 — LOAM

LOAM Feature Extraction
•Edge sharp / flat surfaces
•Curvature metric
•Downsample flat points

LOAM Odometry
Residuals:
•Edge-to-line
•Plane-to-plane
Optimization:

$$\min \sum r_{edge}^2 + \sum r_{plane}^2$$

LOAM Mapping & Global Optimization
•Register to local map
•Re-optimize trajectory
•Loop closure via ICP

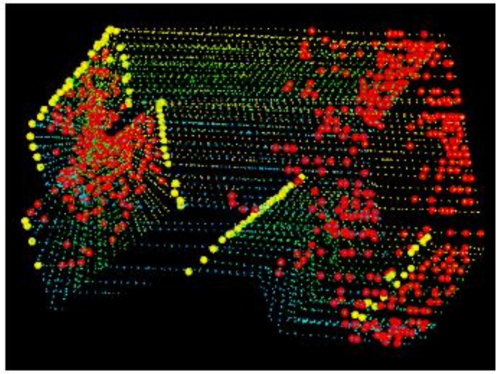


Fig. 5. An example of extracted edge points (yellow) and planar points (red) from lidar cloud taken in a corridor. Meanwhile, the lidar moves toward the wall on the left side of the figure at a speed of 0.5m/s, this results in motion distortion on the wall.

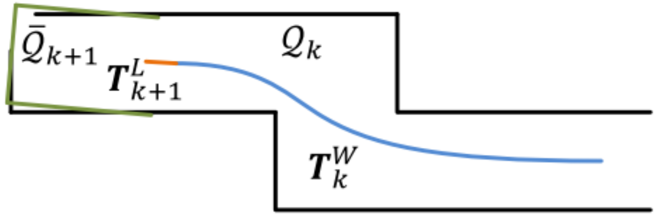
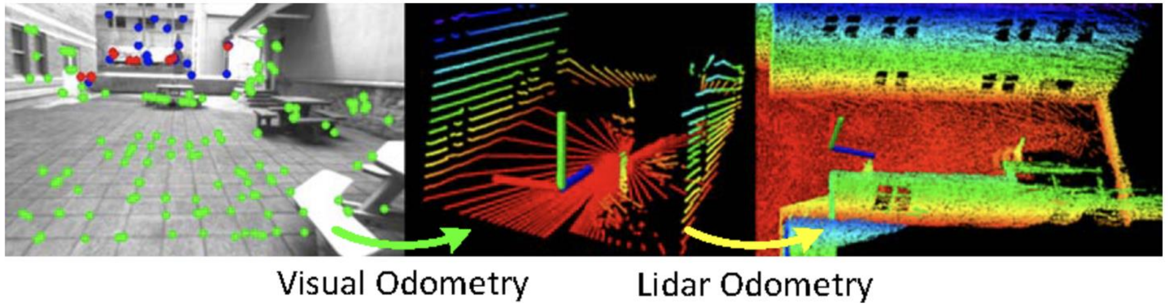


Fig. 8. Illustration of mapping process. The blue colored curve represents the lidar pose on the map, $\boldsymbol{T}_k^W$, generated by the mapping algorithm at sweep k. The orange color curve indicates the lidar motion during sweep $k+1$, $\boldsymbol{T}_{k+1}^L$, computed by the odometry algorithm. With $\boldsymbol{T}_k^W$ and $\boldsymbol{T}_{k+1}^L$, the undistorted point cloud published by the odometry algorithm is projected onto the map, denoted as $\bar{Q}_{k+1}$ (the green colored line segments), and matched with the existing cloud on the map, Q_k (the black colored line segments).

Typical System 2 — V-LOAM

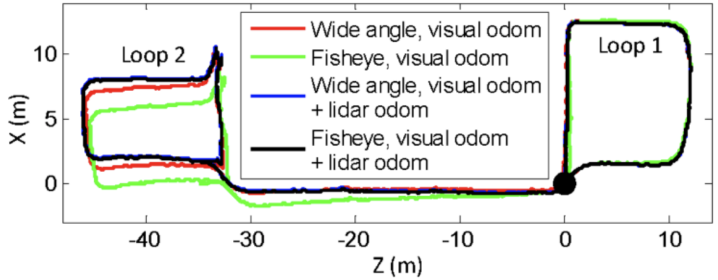
Fuse **vision + LiDAR** to achieve **robust, accurate, and drift-reduced navigation**, especially in feature-sparse scenes where pure LiDAR may fail.



Two-thread architecture (like LOAM)

•**Odometry thread**: Fast VO + LiDAR edge/surf alignment for real-time motion

•**Mapping thread**: LiDAR refinement + keyframe optimization



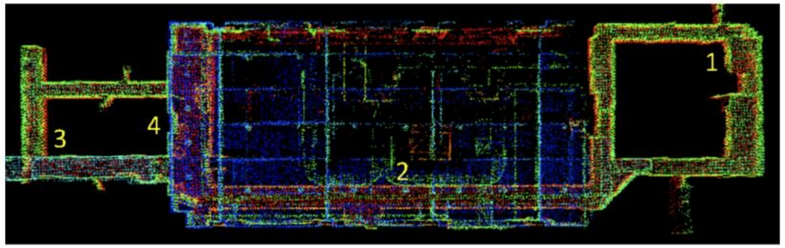
Visual odometry helps initialize LiDAR

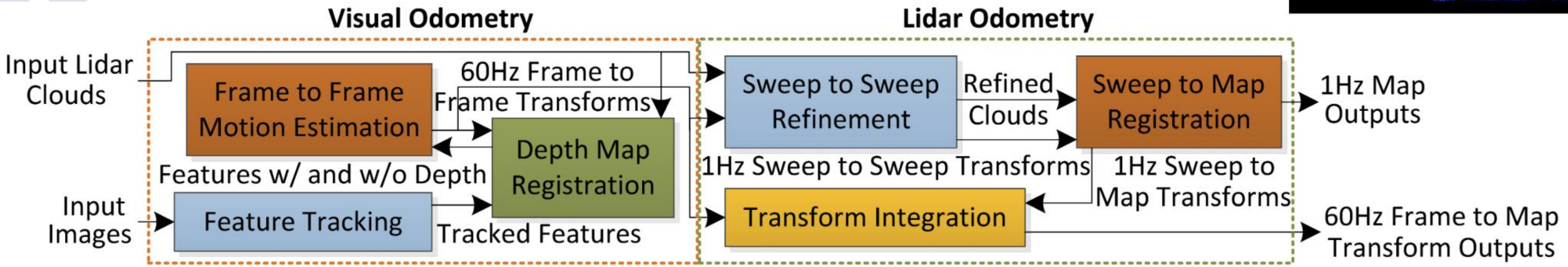
•Improves rotation estimation & robustness in low-geometry environments

Feature fusion

•Point/line visual features + LiDAR edge/plane features

•Joint geometric constraints in motion estimation





System 3 — LIO-SAM

LIO-SAM: Tightly-coupled Lidar Inertial Odometry via Smoothing and Mapping (2020)

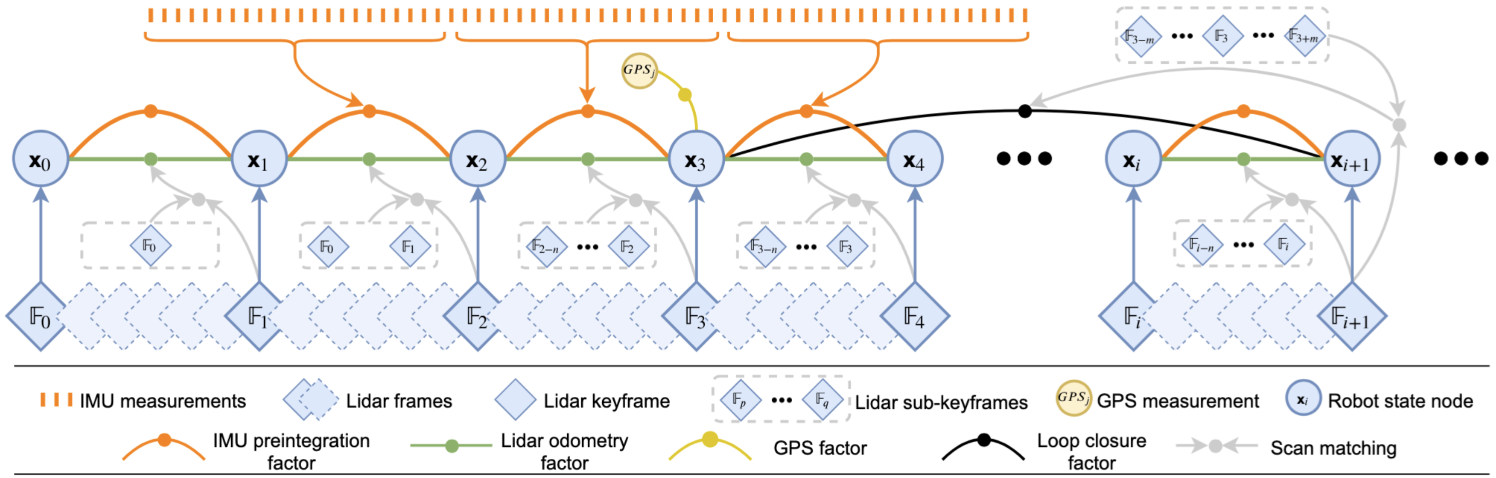
LiDAR + IMU **tightly coupled**
Factor graph optimization (GTSAM)
Loop closure using Scan Context

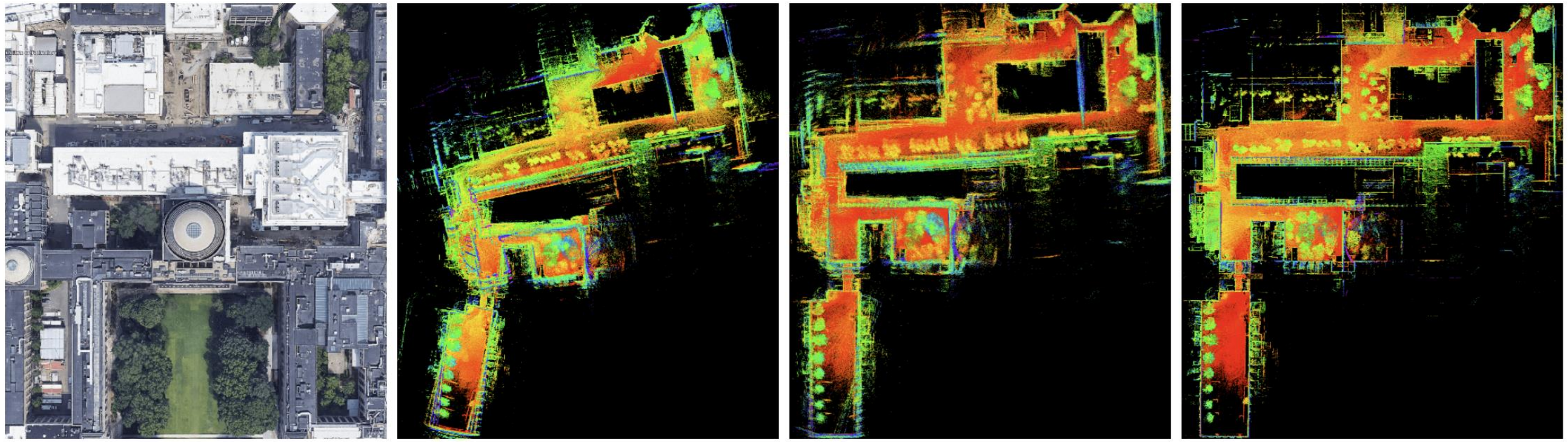
IMU Pre-integration

•Motion prediction prior to LiDAR update
•Helps reduce scan distortion

Factor	Constraint
IMU	Motion prior
LiDAR	Feature residuals
GPS / Loop	Global correction

$$x^* = \arg\min \sum \|r_i(x)\|_{\Sigma_i^{-1}}$$

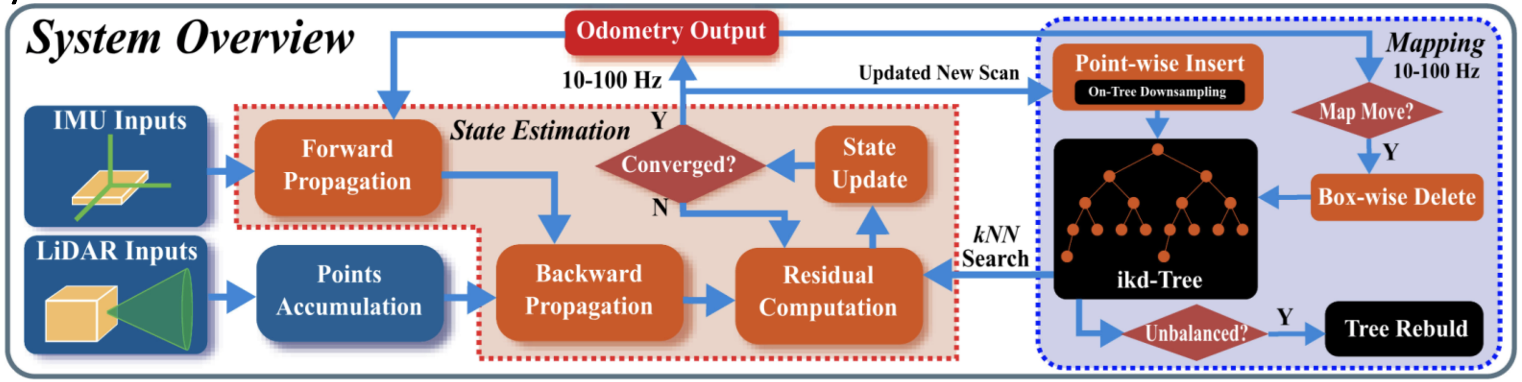




(a) Google Earth (b) LOAM (c) LIOM (d) LIO-SAM

System 3 — FAST-LIO2

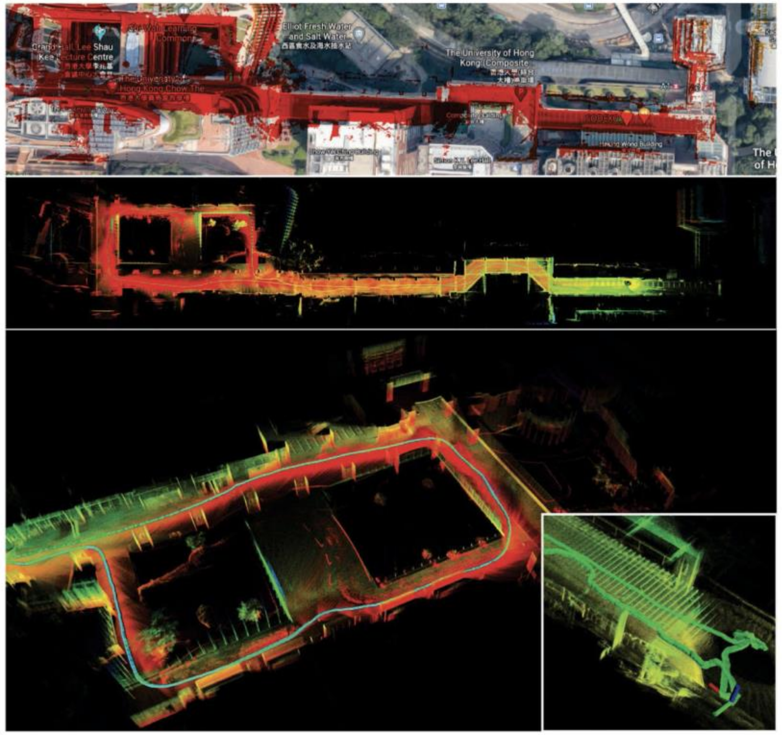
FAST-LIO2: Fast Direct LiDAR-Inertial Odometry



- Fast tightly-coupled LiDAR-IMU odometry
- **ESKF (Error State Kalman Filter)** + **Incremental voxel map**
- Direct raw points → real-time mapping

Incremental Voxel Map

- **IKD-Tree / Voxel hashing**
- Fast nearest-point lookup
- Key improvement vs LOAM/LIO-SAM: **map structure speed**



ESKF Update

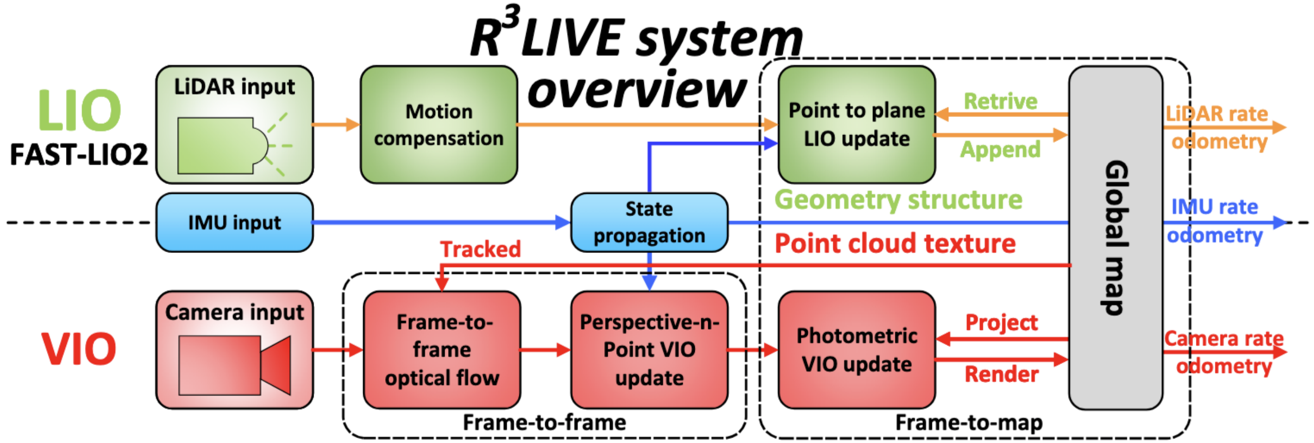
$$x_{k+1} = f(x_k, u_k) + K(z_k - h(x_k))$$

IMU propagates pose
LiDAR updates with point-to-plane residuals

System 4 — R3LIVE (LiDAR + Visual + IMU)

R3LIVE: A Robust, Real-time, RGB-colored, LiDAR-Inertial-Visual tightly-coupled state Estimation and mapping package

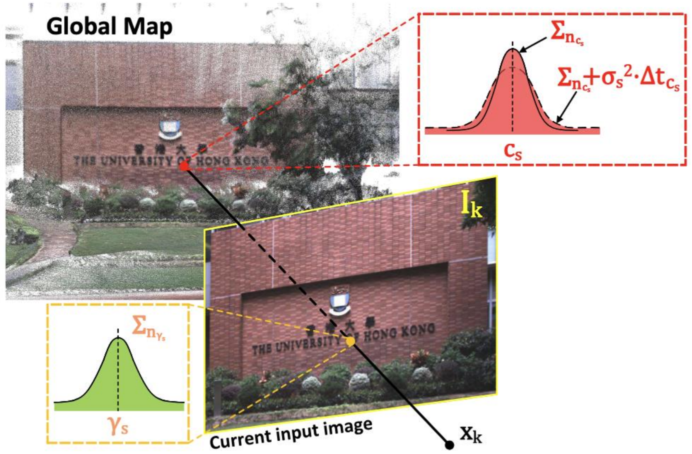
- Dense color mapping (*RGB LiDAR*)
- LiDAR-inertial + **visual BA**
- Joint LiDAR reprojection + image reprojection

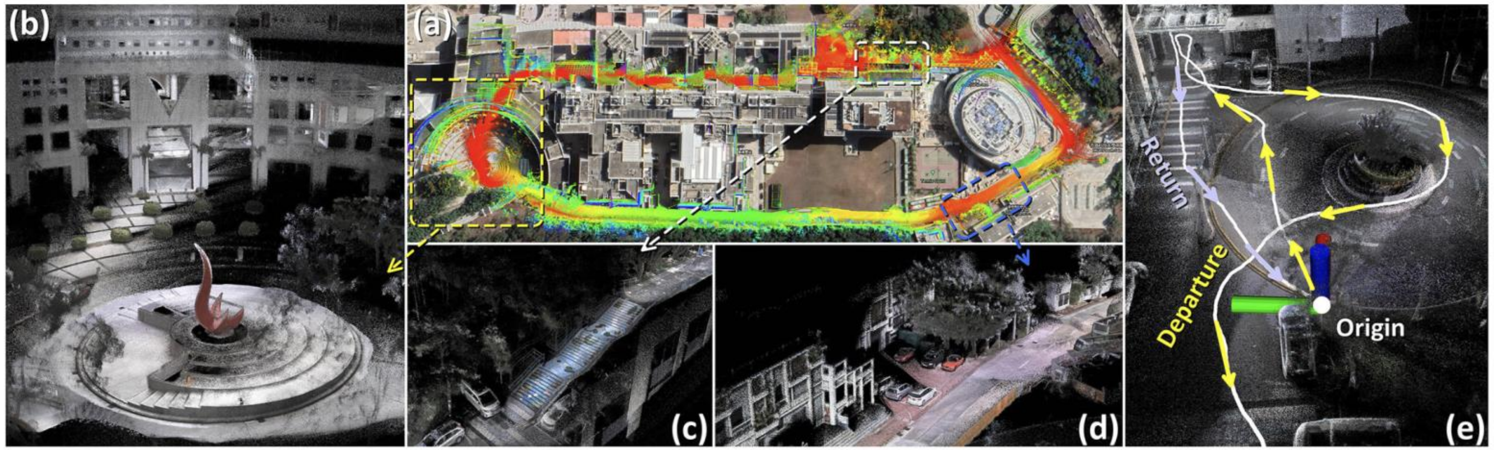


Multi-Modal Residuals

$$r = r_{IMU} + r_{LiDAR} + r_{Visual}$$

- LiDAR geometry
- Pixel reprojection error
- Photometric consistency



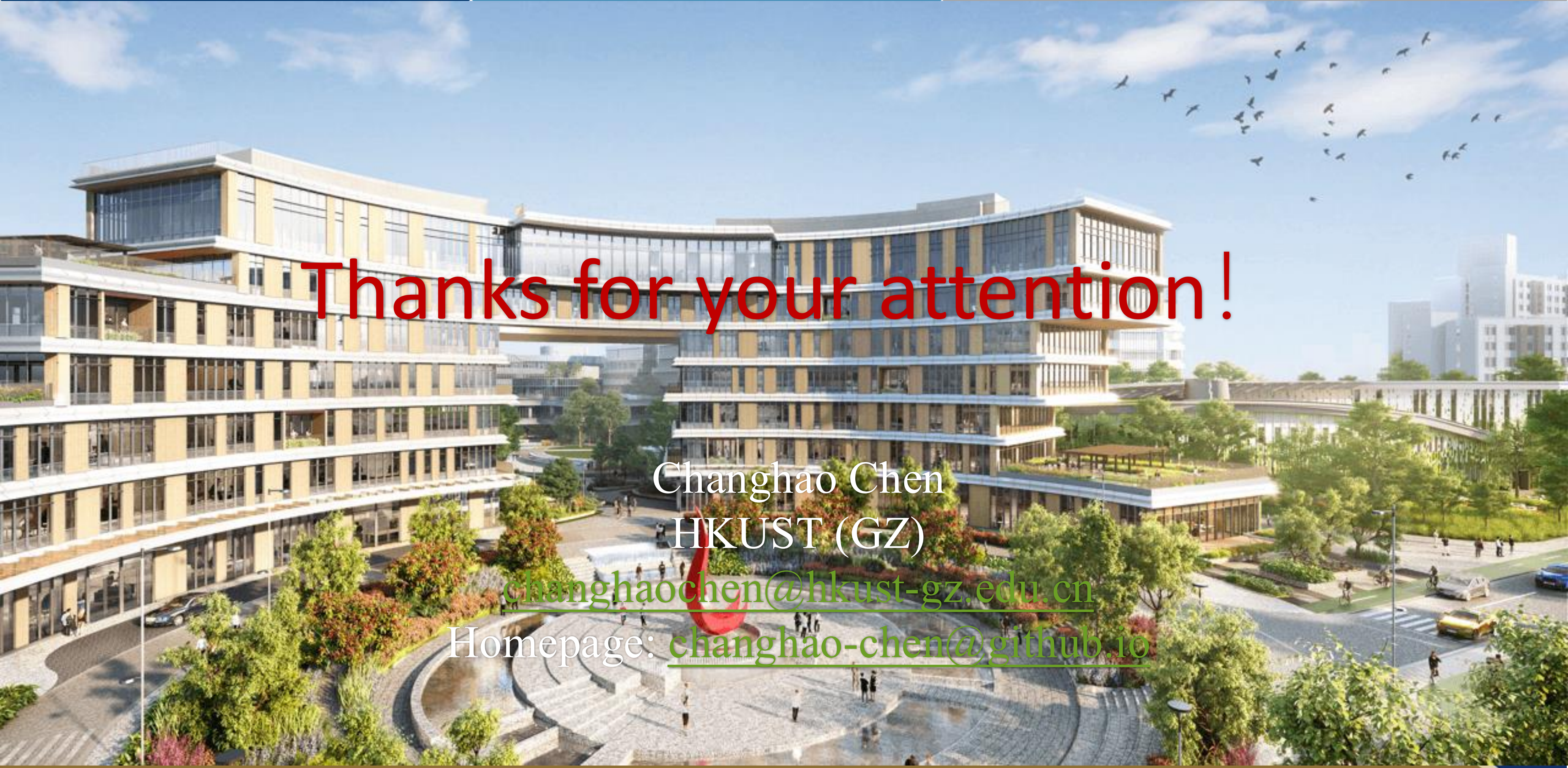


Comparison Summary

System	Sensors	Optimization	Strengths	Weakness
LOAM	LiDAR	Feature ICP	Simple, accurate	No IMU
LIO-SAM	LiDAR+IMU	Factor graph	Robust, loop closure	Heavier
FAST-LIO2	LiDAR+IMU	ESKF	Fastest mapping, real-time	No visual
R3LIVE	LiDAR+Cam+IMU	Joint BA	Dense colored map	Heavy compute

Research Trends

- Neural SLAM (NeRF-SLAM, NICE-SLAM)
- BEV multimodal fusion
- Event camera + LiDAR fusion
- Self-supervised LiDAR-Visual localization and Mapping
- Dynamic scene modeling



Thanks for your attention！

Changhao Chen
HKUST (GZ)
changhaochen@hkust-gz.edu.cn
Homepage: changhao-chen@github.io